



**Verified Carbon
Standard**

THE CLEAN CHOOOLA INITIATIVE

Project title	The Clean Choola Initiative
Project ID	Verra Project ID
Monitoring period	15-March-2023 to 31-October-2025
Crediting period	15-March-2023 to 14-March-2030
Original date of issue	For pipeline listing, 03-December-2024 is the date of submission
Most recent date of issue	03-December-2024 is the date on which the document was most recently submitted
Version	01
VCS Standard Version	4.7
Prepared by	Core Carbon Canada

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Audit History.....	6
1.3	Sectoral Scope and Project Type	6
1.4	Project Eligibility	6
1.5	Project Design	7
1.6	Project Proponent	10
1.7	Other Entities Involved in the Project	11
1.8	Ownership.....	11
1.9	Project Start Date	11
1.10	Project Crediting Period	11
1.11	Project Scale and Estimated GHG Emission Reductions or Removals	12
1.12	Description of the Project Activity	12
1.13	Project Location	15
1.14	Conditions Prior to Project Initiation	15
1.15	Compliance with Laws, Statutes and Other Regulatory Frameworks.....	16
1.16	Double Counting and Participation under Other GHG Programs	16
1.17	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	17
1.18	Sustainable Development Contributions	18
1.19	Additional Information Relevant to the Project	24
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	24
2.1	Stakeholder Engagement and Consultation.....	24
2.2	Risks to Stakeholders and the Environment.....	28
2.3	Respect for Human Rights and Equity	30
2.4	Ecosystem Health	33
3	APPLICATION OF METHODOLOGY.....	35
3.1	Title and Reference of Methodology	35
3.2	Applicability of Methodology	35
3.3	Project Boundary	42
3.4	Baseline Scenario	44

3.5	Additionality	44
3.6	Methodology Deviations	46
4	IMPLEMENTATION STATUS	46
4.1	Implementation Status of the Project Activity	46
5	QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	46
5.1	Baseline Emissions	46
5.2	Project Emissions	47
5.3	Leakage Emissions	47
5.4	Estimated GHG Emission Reductions and Carbon Dioxide Removals	47
6	MONITORING	49
6.1	Data and Parameters Available at Validation	49
6.2	Data and Parameters Monitored.....	52
6.3	Monitoring Plan.....	57
7	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	60
7.1	Data and Parameters Monitored.....	60
7.2	Baseline Emissions	61
7.3	Project Emissions	61
7.4	Leakage Emissions	61
7.5	GHG Emission Reductions and Carbon Dioxide Removals	62
	APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION.....	63
	APPENDIX X: <TITLE OF APPENDIX>	64

1 PROJECT DETAILS

1.1 Summary Description of the Project

"The Clean Choola Initiative" aims to distribute improved cookstoves to households mostly dependent on traditional chulah in Pakistan. The project is being implemented by Core Carbon Canada, this project seeks to improve cooking practices and enhance livelihoods within rural households. (hereafter "The Clean Choola Initiative" is called as "The Project")

This project represents a transformative initiative aimed at changing from traditional cooking stoves to improved cookstoves, promoting hygiene, and empowering rural communities, particularly women, from all over the Globe. At its core is the introduction of the double-stove innovation, offering a cost-effective and environmentally friendly alternative to traditional single-stove setups. Equipped with a chimney for improved ventilation, the project minimizes cooking time and smoke emissions, significantly improving the cooking experience and promoting better respiratory and eye health for women disproportionately affected by indoor air pollution.

Elevated on raised earthen platforms, these stoves provide clean dining spaces fueled by readily available agricultural waste or cow dung, reducing reliance on biomass and saving time for households. Adding a designated washing area shows the project cares about keeping families healthy by promoting good hygiene habits.

The project's impact extends beyond practical cooking solutions. Through the training of Barefoot Entrepreneurs, known as "Chulah or Stove Sisters," rural women/housewives have the skills to construct and maintain these stoves.

Overall, "The Project" represents a holistic approach to addressing cooking challenges in rural areas across the Globe, initiated in Pakistan and further expanding to other developing countries and least-developed countries, combining innovation, community empowerment, and sustainability to improve the lives of countless families while fostering a sense of dignity and self-sufficiency within communities.

The location of the project

The improved cookstove project is initiated in Pakistan with the GPS coordinates below.

Location	Latitudes	Longitude
Pakistan	30.3753° N	69.3451° E



An explanation of how the project is expected to generate GHG emission reductions or removals.

The "Clean Choola Initiative" project is expected to significantly reduce greenhouse gas (GHG) emissions by implementing improved cookstoves. These stoves will replace traditional open-flame cooking methods with more efficient double-stove innovations, thus minimizing using biomass fuels like wood or agricultural waste. The improved efficiency of these Improved Cookstoves (ICS) ensures better thermal transfer to cooking pots, reducing fuel consumption. Moreover, adding a chimney for improved ventilation facilitates complete combustion, decreasing harmful pollutants and GHG emissions. The project aims to substantially reduce GHG emissions in rural households by promoting cleaner and more efficient cooking practices.

Construction and Commissioning Period: The construction and commissioning of the 50,000 ICS took place from 15-March-2023 to 31-October-2025. During this period, the cookstoves were built, tested, and deployed to ensure they met the required standards and were fully operational.

Operation Period: The continued operation of these cookstoves began immediately after commissioning on 15-March-2023 and will be rigorously monitored to ensure consistent performance and emission reductions. The operation period involves regular maintenance,

user training, and performance assessments to ensure the cookstoves function efficiently and reduce GHG emissions.

A brief description of the baseline scenario existed before the project's implementation.

The baseline scenario is the use of non-renewable firewood by the target households to meet the thermal energy needs provided by project cookstoves in the absence of project activity.

An estimate of annual average and total reductions and removals.

During the registration of the project activity, one instance involved the construction of 50,000 cookstoves for Project Instance 1. The average annual GHG emission reduction from this first project instance is 281,609 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/	15-March-2023 to 14-March-2030	VCS	To be decided (TBD)	7
Verification	15-March-2023 to 31-October-2025		TBD	2

1.3 Sectoral Scope and Project Type

Sectoral scope¹	Scope 03: Energy Demand
Project activity type	Type I- Renewable energy projects

Sectoral scope	Not Applicable
AFOLU project category²	Not Applicable
Project activity type	Not Applicable

1.4 Project Eligibility

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² See Appendix 1 of the VCS Standard

1.4.1 General eligibility

As per section 2.1 of the scope of the VCS program, VCS standard V 4.7, the scope of the VCS program is;

- 1) The six Kyoto Protocol greenhouse gases.
- 2) Ozone-depleting substances.
- 3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process.
- 4) Project activities supported by a methodology approved under an approved GHG program unless explicitly excluded (see the Verra website for exclusions).
- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.

The grouped project activity involves the construction of improved cookstoves in households/communities, thereby reducing household firewood consumption. This project activity falls under the efficiency improvements in thermal applications; Therefore, this project activity is eligible under VCS standard version 4.7. Thus, GHG emission reductions can be achieved. The project is eligible under the scope (the six Kyoto Protocol greenhouse gases) of the VCS Program; therefore, it is eligible under the scope of the VCS Program.

The project activity falls under sectoral scope 03: Energy Demand," and is not part of any categories falling under exclusion criteria in Table 2.1 of the VCS standard V4.7. The project involves CO₂ emission reduction, a covered GHG under the VCS programme. Therefore, the project is eligible under the scope of the VCS program and in line with the VCS Standard V4.7.

1.4.2 AFOLU project eligibility

The project is not an AFOLU project; thus, this section is not applicable.

1.4.3 Transfer project eligibility

Not Applicable

1.5 Project Design

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

1.5.1 Grouped Project Design

S.No	Eligibility Criteria	Description	How the PI 1 is complying and Evidence
1.	VCS standard version 4.7 section 3.6.16 Meets the applicability conditions set out in the methodology applied to the project.	The Improved Cooking Stoves (ICS) project meets the requirements set out in Section 3.2, as the product would be used in rural households to enhance the thermal efficiency of the cookstoves by replacing it with the ICS. The deployed ICS will have a minimum 25% thermal efficiency.	Please see section 3.2 of this document for details.
2.	Use the technologies or measures specified in the project description.	The project activity would deploy only Improved cookstoves (ICS) in the household by replacing traditional cookstoves.	The ICS have higher efficiency and better emission parameters than the baseline scenario. Evidence: Manufacturer's specification/installation records
3.	Apply the technologies or measures in the same manner as specified in the project description.	The project activity would deploy only Improved cookstoves (ICS) in the household by replacing traditional cookstoves.	The ICS has higher efficiency and better emission parameters than the baseline scenario. Evidence: Manufacturer's specification/installation records
4.	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	The grouped project ICS initially constructs the cookstoves in Pakistan and will further expand the project to other developing countries and least-developed countries. Only subject to the same baseline scenario determined in Section 3.4.	The PI has the same baseline scenario defined in section 3.4 of this document, i.e., using traditional three-stone cookstoves for cooking in different regions of Pakistan and other developing countries and LDC.
5.	Have characteristics with respect to additionality that are consistent with the	All new project activity instances will use the activity method to demonstrate additionality. Step 1: Regulatory Surplus There is	The PI 1 is additional and is described in section 3.5 of this document.

	initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances.	no mandated government programme or policy in the host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List Including new project activity instances will comply with the positive list as it satisfies criterion 1, which meets all the methodology's applicability conditions.	
--	---	---	--

Along with the above points, the Inclusion of New Project Activity Instances follows the criteria below.

Grouped projects include new project activity instances after the initial validation of the project. New project activity instances shall:

S.No	Criterion	How do the new project activity instances comply
1	VCS standard version 4.7 section 3.6.17 Occurs within one of the designated geographic areas specified in the project description.	New project activity instances (Energy Efficient Cook Stoves) will occur in the designated geographic areas specified in the project description.
2	Comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	New project activity instances will comply with the eligibility criteria mentioned above to include new project activity instances.
3	Be included in the monitoring report with sufficient technical, financial, geographic and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	Information on new project activity instances is to be included in the monitoring report, along with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable

		eligibility criteria and enable sampling by the validation/verification body.
4	Be validated against the applicable set of eligibility criteria at the time of verification. Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions)	This document mentions that new project activity instances should be included by validating eligibility criteria. Ownership of the new project activity instance needs to be evidenced from the start date of the respective project activity instance.
5	Have a start date that is the same as or later than the grouped project start date.	The start date of new project activity instances should be after the start date of the grouped project activity.
6	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period and new instances are eligible for crediting from the start of the next verification period.	A new project activity instance can claim credits from the crediting period start date to the end of the project crediting period.
7	Not be or have been enrolled in another VCS project.	The new project activity instance has not been enrolled under another VCS project.

1.6 Project Proponent

Organization name	Core Carbon Canada
Contact person	Mr. Arif Alauddin
Title	Director
Address	25-5230 Glen Erin Drive, Mississauga, ON L5M5Z7, Canada.
Telephone	+1 437-994-8914 +923008548914
Email	arifalauddin@corecarbon.ca

1.7 Other Entities Involved in the Project

Organization name	Not Applicable
Role in the project	
Contact person	
Title	
Address	
Telephone	
Email	

1.8 Ownership

In all the project activity instances under the project activity, Core Carbon Canada will implement the project devices (Energy Efficient Improved Cookstoves) through implementing partners. Beneficiaries will be considered legal owners of the project devices. In this mechanism, energy-efficient cookstoves will be constructed for beneficiaries. The ownership rights of the carbon assets generated from this project will be with Core Carbon Canada, which will sign a detailed End-user beneficiary agreement mentioning all the criteria with all beneficiaries before the commissioning of the project devices.

1.9 Project Start Date

Project start date	15-March-2023
Justification	This date corresponds with the first ICS installation.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
-------------------------	---

Start and end date of first or fixed crediting period	15-March-2023 to 14-March-2030
--	--------------------------------

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

- ☐ < 300,000 tCO₂e/year (project)
- ☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
15-March-2023 to 31-December-2023	224,515
01-January-2024 to 31-December-2024	281,609
01-January-2025 to 31-December-2025	281,609
01-January-2026 to 31-December-2026	281,609
01-January-2027 to 31-December-2027	281,609
01-January-2028 to 31-December-2028	281,609
01-January-2029 to 31-December-2029	281,609
01-January-2030 to 14-March-2030	55,550
Total estimated ERRs during the first or fixed crediting period	19,69,719
Total number of years	07
Average annual ERRs	281,609

1.12 Description of the Project Activity

The "Clean Choola Initiative" is an improved cooking stove that is elevated on a lime-stabilized earthen platform, designed to withstand various environmental challenges such

as rain, floodwaters, and earthquakes, making it disaster risk reduction (DRR)-compliant. Constructed using techniques drawn from the vernacular traditions of Pakistan, the stove is built by rural women skilled in earthen construction methods. The structure is covered with a plaster mixed with straw and cow dung, providing elasticity and smoothness to the surface. Adding lime strengthens the structure and prevents the typical disintegration of earth surfaces, drawing from Pakistan's ancient heritage dating back to the 16th century.

The double-stove design on earthen platforms ensures a hygienic cooking environment by preventing contamination from animal droppings and dirt commonly found in floor-mounted open-fire stoves. An air regulation pipe facilitates heat transfer from the combustion chamber to the secondary chamber, allowing cooking on both stoves with fuel used only in one. The incorporated chimney emits smoke only when there is a fire.

Reaches a specific temperature, reducing indoor air pollution. The platform provides a clean space for gathering or sitting during meals. The washing area promotes regular handwashing before cooking and eating. The double stove can hold two pots, reducing overall cooking time. Wood gasification combustion techniques enable longer fire burning and less use of expensive wood fuel. The average life span of an improved cookstove is around ten years, and efficiency is 34.8%.

From a social perspective, the Pakistan Chulah provides a clean dining and socializing space for families, elevating the status of mothers within the household and community. It also serves as an expression of the housewife's creativity in its construction and design. The use of layered lime-mud in construction and lime-mud coating protects from damage due to floods or rain, ensuring the longevity and durability of the structure.



Children watching their lunch being cooked on a clean earthen platform



Stages of construction

Stage I: Marking Foundations	- Place four pegs on the corners to mark the foundation. - Excavate to a depth of 12 inches and fill with a 4-inch-thick layer of lime concrete.
Stage II: Constricting Raised Platform	- Mark a platform using 1:6 lime and mud layers. - Raise the platform to a height of at least 15 inches to create a base for the double stove arrangement. - The raised platform provides a hygienic cooking space.
Stage III: Layout for Double Stove	- Lay out the location of the two stoves and chimney by placing pegs. - Construct steps for access to the platform.
Stage IV: Construction of Double Stove	- Build the double stove, ensuring proper openings and heat movement. - Construct the chimney to a height that provides smoke protection.
Stage V: Adding Handwashing Arrangement	- Attach a platform for handwashing to the main platform. - Include a slightly sunken sink to drain wastewater to water a plantation bed, avoiding sewage.
Stage VI: Completion	- Complete the rear storage wall with layered mud or mud bricks. - Plaster the entire structure using a lime/mud/straw mix for water protection.



1.13 Project Location

The project is initially implemented in Pakistan with coordinates of 30.3753° N latitudes and 69.3451° E longitudes.



Geographical location of Pakistan

1.14 Conditions Prior to Project Initiation

The project aims to decrease the reliance on fuel wood for cooking in rural households. Many households in these areas of Pakistan and other developing and least-developed countries use traditional stoves and open fires, leading to inefficient firewood burning and contributing to greenhouse gas emissions and deforestation. The project aims to reduce emissions and conserve forests by introducing improved cooking stoves. The baseline scenario reflects the

ongoing use of traditional cookstoves in rural households. Section 3.4 (Baseline Scenario) for a detailed description.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

National Action Plan: Sustainable Energy For All³ Pakistan - Advancing Implementation of Pakistan's National Clean Air Policy Through Clean Cooking Fuels and Technologies [PK-23-003]⁴

National Climate Change Policy⁵ are some of the laws in Pakistan that do not explicitly mandate the use of improved cookstoves in households. While the government encourages the use of clean fuels such as LPG, many families face challenges in affording and refilling LPG cylinders due to limited income. As a result, the usage of LPG remains low, and most rural households continue to rely on conventional stoves for cooking. Notably, there is no explicit mention of improved cookstove projects in existing laws and regulations. Therefore, the implementation of such projects is a voluntary initiative by project proponents.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and the date of project inactivity under the other GHG program.

Is the project active under the other program?

☐ Yes ☒ No

Project proponents, or their authorized representative, must attest that the project is no longer active in the other GHG program in the Registration Representation.

³ National Action Plan: Sustainable Energy For All | United Nations Development Programme (undp.org)

⁴ Pakistan - Advancing Implementation of Pakistan's National Clean Air Policy Through Clean Cooking Fuels and Technologies [PK-23-003] | Climate & Clean Air Coalition (ccacoalition.org)

⁵ http://www.gcisc.org.pk/National_Climate_Change_Policy_2012.pdf

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

If yes, provide the program name(s), reason(s) and date for the rejection, justification of eligibility under the VCS Program, and any other relevant information.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the VCS Program Definitions for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes

☒ No

If yes to all:

Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

As per section 3.17 of the VCS Program Standard, v4.7 requires to demonstrate to the sustainable development in the project area as defined in the UN Sustainable Development Goals.

The project activity is installed in rural areas of Pakistan. The PP with the help of local partners, and the field team implemented the project activity. Furthermore, the PP (stove sisters) is running awareness programmes to inform people about the value of utilizing an energy-efficient improved cookstove, lowering indoor pollution, and their part in the decrease in greenhouse gas emissions. Customers' quality of life is improved, particularly that of women and children, when fuel wood consumption is reduced because less time is needed for cooking and wood gathering. Because of the project work throughout this monitoring period, the PP has identified and monitored below key SDGs. The following contributions are explained:

SDG 1: SDG 1 focuses on the number of households that received energy-efficient improved cookstoves, contributing to increased access to clean energy. Under the project activity, 50,000 energy-efficient cookstoves (Instance 1) were installed with the support of the Heritage Foundation. This initiative enhances access to clean energy and contributes to SDG Indicator 1.4: Proportion of the population living in households with access to clean cooking solutions and technology.

SDG 3: SDG 3 deals with a number of individuals experiencing health benefits due to reduced exposure to indoor air pollution, PP has installed energy-efficient improved cookstoves of 50,000 households which is eliminating their exposure to hazardous gases such as CO, and

PM2.5 released from the inefficient burning of fuel wood. This would avoid premature deaths and chronic diseases in rural areas of Pakistan where access to public and private health care is limited. Therefore, the project activity is contributing towards the well-being of the local people. This would reduce exposure to smoke and Particulate matter (PM) emissions.

SDG 4: SDG 4 deals with the number of individuals who received any informal training. The project activity contributed to the increasing vocational and relevant skills of at least 100 local individuals, with a focus on targeting youth and women by providing non-formal education and training on issues related to climate change, the purpose of installing the energy-efficient improved cookstoves, with specific skill building on cookstove construction, data collections, and surveying activities related to cookstove installation and its monitoring activities.

SDG 5: The SDG5 talks about achieving gender equality and empowering all women and girls. Reduce the amount of time that women and children must spend gathering and transporting firewood from distant trees, as well as the amount of time that they must spend cooking over hazardous, smoke-filled open fires, by using the ICS, which uses less fuel wood. The project's activities will address gender disparity in rural areas of Pakistan communities by giving women and children the opportunity to participate in more productive activities. These duties are a key cause of gender inequality in these communities.

SDG 7: The SDG7 deals with ensuring access to affordable, reliable, sustainable, and modern energy for all by 2030. As previously mentioned, the main activity under PI1 and PI2 is the installation of clean, improved project stoves for low-income households that meet their cooking needs primarily with wood fuel and cannot purchase clean, improved cookstoves.

SDG 8: The UN has called for promoting full and productive employment and decent work for all, by 2030 in addition to sustained, inclusive and sustainable economic growth. To do this, policies that support productive activities, the creation of good jobs, entrepreneurship, creativity, and innovation, among other things, must be promoted. In terms of the project, providing ICS to low-income homes requires administrative assistance for installation, continued operation, and monitoring, as well as qualified staff. the project's backing, the public dissemination of its technology, and the expansion of its activities into new regions. All of this is helping to create important and fruitful jobs, particularly for young people in the area. PP has employed 150 local youth in direct employment and 100 in indirect employment.

SDG 13: The UN in' 2015 called for urgent action to combat climate change as a result of human-induced activities and its impacts, which is now resulting in more and irreversible

damage to humanity and the planet every year. The project aims to reduce the consumption of fuel wood for cooking purposes which is majorly sourced from the forest areas. As a result, the project activity was achieved to reduce greenhouse gas emissions significantly over the monitoring period.

SDG 15: SDG 15 deals with the progress towards sustainable forest management by increasing above-ground biomass in forests. Contribute an estimated reduction of deforestation of 4.81 tons of woody biomass, per stove, per year, from forests surrounding the communities and reducing pressure on forest reserves.

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	1.4	Project Specific Indicator: 1.4.1 Proportion of the population living in households with access to clean fuel and technology.	Implement activities to increase.	The project activity has installed 50,000 energy-efficient cookstoves to households primarily relying on traditional cookstoves and increased access to clean fuel and technology. According to the monitoring survey conducted by the PP using a random sampling approach, it was found that the entire population, 100%, is utilizing the ICS.	Increase access to clean cooking technology with project stove installations is approximately 1 million households under the project lifecycle.
2)	3.9	Project Specific Indicator: Number of individuals experiencing health benefits due to reduced exposure to indoor air pollution.	Implement activities to increase.	100% of beneficiaries have reported that they are witnessing improvement in indoor air quality due to smokeless improved cookstoves and experienced health benefits.	100% of beneficiaries have reported that they are witnessing improvement in indoor air quality due to smokeless improved cookstoves and experienced health benefits.

3)	4.3	Project Specific Indicator: Number of individuals who received any informal training	Implement activities to increase.	The project activity contributed to the increasing vocational and relevant skills of at least 100 local individuals ("Chulah or Stove Sisters."), with a focus on targeting youth and women by providing non-formal education and training on issues related to climate change, the purpose of installing the energy-efficient improved cookstoves, with specific skill building on cookstove construction, data collections, and surveying activities related to cookstove installation and it's monitoring activities.	Contribute to increasing vocational and relevant skills of at least 500 local individuals (with a focus on targeting women and youth) by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to stove installation and its monitoring under VCS
4)	5.4	Project Specific Indicator: Average number of hours saved per household due to the use of the project device.	Implement activities to decrease.	According to the methodology "VM0050 Time Savings from Improved Cookstoves," which uses a default value of 0.5 hours per household per day for each cookstove, the distribution of cookstoves to 50,000 households' results in a total time savings of approximately 25,000 hours per day.	The project activity is expected to distribute 1 million cookstoves and expected to save 0.5 million hours/day over the lifetime of the project.
5)	7.1	7.1.2 Proportion of population with primary reliance on clean fuels and technology	Implement activities to increase.	Increasing access to clean cooking technology by installing a set of project stoves in 50,000 households. As per the Pakistan chulah monitoring survey carried out by the PP as per the random sampling approach 100% population is using the ICS.	Increase access to clean cooking technology with project stove installations in approximately 1 million households under the project lifecycle.

6)	8.3	Project Specific Indicator: Number of employment generation by the project activity.	Implement activities to increase.	Contributed to the generation of employment in the informal sector (total economy, agriculture, and non-agriculture) by contracting locals with 100	Contribute to the generation of employment in the informal sector (total economy, agriculture, and non-agriculture) by contracting locals with targeted employment of 250 employees for varying lengths of time over the project lifetime with a focus on hiring females.
7)	13.0	Project Specific indicator: Tonnes of greenhouse gas emission avoided.	Implement activities to increase.	The total emission reduction in the current monitoring period is 5,61,674 tCO ₂ eq.	Contribute to an estimated annual reduction of 281,609 tonnes of CO ₂ equivalent greenhouse gas emissions.
8)	15.2	Project Specific Indicator: Fuelwood saved by project device	Implemented activities to increase	"In the baseline scenario, households were using 4.81 tons of non-renewable biomass per year. In the project scenario, this non-renewable biomass is completely replaced with renewable biomass, leading to a 100% reduction in the use of non-renewable biomass."	In the baseline scenario, households were using 4.81 tons of non-renewable biomass per year. Over a seven-year period, this amounts to approximately 33.67 tons of non-renewable biomass. In the project scenario, this non-renewable biomass is entirely replaced with renewable biomass, resulting in a 100% reduction in non-renewable biomass consumption. Consequently, over the seven-year crediting period, this transition saves approximately 33.67 tons of non-renewable biomass.

1.19 Additional Information Relevant to the Project

Leakage Management

This is not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No other information is available at this stage.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The PP identified the primary stakeholders who are informed and consulted about the project, including cookstove beneficiaries, groups, or communities that:

- Are impacted or likely to be affected by the project (project-affected parties); and
- May have a stake in the project (other interested parties)

The project involves installing energy-efficient cookstoves, categorizing the following parties as project-affected parties and other interested parties:

Project-affected parties: End users (Cookstove beneficiaries)

Other Interested Parties: ICS implementers and their representatives, Local Women Groups and NGOs, Village

	Heads, block-level officials, and National Government Officials.
Legal or customary tenure/access rights	The project activity involves the construction of energy-efficient, improved cookstoves and does not involve any legal or customary tenure/access rights. The target end users are legal owners of ICS. Full and uncontested legal ownership of the carbon credits generated will be with PP, for which an appropriate agreement will be signed with each beneficiary.
Stakeholder diversity and changes over time	There is no change expected in the make-up of stakeholders over time.
Expected changes in well-being	Expected changes in the well-being of the stakeholders are as follows: • The energy-efficient project ICS will reduce the emission of indoor pollutants, leading to health benefits for the beneficiaries. • Project ICS shall save time in cooking activities and fuel procurement; the time saved can be used for other productive and income-generating activities. • Money will be saved as the project ICS requires less fuel. The project will create local employment opportunities in operational and management roles as well as in the implementation and logistics of ICS.
Location of stakeholders	The geographical boundary of the project is initiated in Pakistan and will further expand to developing and least-developed countries. The geographical location of the project activity instance is Pakistan; hence, the stakeholders belong to the villages under the state of Sindh, Pakistan. The project impacts no areas outside the project boundary.
Location of resources	The project is the installation and deployment of new ICS to the participating households and does not involve resources owned by the stakeholders.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	27-January -2024
Stakeholder engagement process	The process of engaging stakeholders in a culturally appropriate manner involved several steps. Firstly, personal interactions were prioritized, recognizing the importance of direct communication. Invitations were tailored to different segments of stakeholders. Due to the high level of illiteracy and limited access to modern communication tools among beneficiaries and the local population, unconventional methods were utilized. Specifically, the Heritage Foundation's Eight "Brigades" were mobilized to disseminate information in rural Sindh. Additionally, various non-conventional methods of invitation were employed to ensure inclusivity. Formal letters were sent to governmental bodies, including the Energy Department and Health Department of the Government of Sindh, district administration and NGOs operating in the area. Invitations were extended through multiple channels, such as WhatsApp groups, local universities, and media outlets. Direct outreach to politicians at all levels, facilitated by organizations like the USAID/EU-supported Association for Development of Local Governance and LEAD for SDGs Program, further enhanced the invitation process. It was emphasized that invitations should be widely shared to maximize participation and foster inclusivity in the event.
Consultation outcome	The project proponent discussed that Switching to improved cookstoves (ICS) presents a significant improvement in household cooking practices. With ICS, families can enjoy cleaner and more efficient cooking, as these stoves produce less smoke and particulate matter than traditional ones. Beyond the health benefits of reduced indoor air pollution, ICS contributes to environmental sustainability by conserving fuelwood and reducing greenhouse gas emissions. Additionally, using ICS saves time and effort for households, particularly for women and children, who are often responsible for cooking and fuel collection. Overall, adopting ICS enhances the well-being of households and the environment, making it a valuable investment in sustainable living.

Ongoing communication	Stakeholder meetings will be held continuously, and information about physical, postal, and electronic addresses will be provided for posting any grievances or suggestions related to the project to improve the project design and implementation continuously.
Stakeholder input	Comments received in stakeholder meetings are positive and do not impact the project.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project proponent obtained approval from the Village Heads through documented consultation meetings before initiating construction in the villages. Additionally, installing energy-efficient improved cookstoves is voluntary, allowing households in participating villages to decide whether to participate. The improved cookstoves are only constructed for households willing to use them. As of the current monitoring period, there are no ongoing or unresolved conflicts related to the project.
Outcome of FPIC	The project involves building energy-efficient improved cookstoves and does not require any land acquisition. The installation of these cookstoves occurs at the residences of project beneficiaries. The field team only visits beneficiary households with the homeowner's permission. The project proponent ensures that all property rights are acknowledged, respected, and upheld. Furthermore, the project has not relocated individuals without their consent, encroached on land, or caused any forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	Stakeholders register grievances through the Continuous Input/Grievance Register, maintained at the project proponent's office. A dedicated phone number and email address are also provided for stakeholders to communicate complaints or grievances. Stakeholders can also directly approach field coordinators or stove sisters to address
----------------------------	---

	concerns. Grievances will be addressed within 30 days of being registered.
Grievance redress procedure	The grievance redress procedures offer various channels for stakeholders to submit complaints, including dedicated registers, hotlines, email, and direct contact with project staff, ensuring accessibility and inclusivity. Upon receipt, grievances undergo a structured process involving acknowledgement thorough investigation, and transparent communication about progress and outcomes. Efforts are made to resolve issues promptly and fairly, with consideration for culturally sensitive conflict resolution. Stakeholders remain informed and engaged throughout, with opportunities for further input and feedback to improve the process continually.

Grievances received	Resolution and outcome
There were no grievances raised during the current monitoring period	As there were no grievances raised, no resolutions were taken into account.

2.1.5 Public Comments

Comments will be added after the public comment period is over.

Comments received	Actions taken

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Core Canada management has vast experience in developing the carbon projects.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	By implementing this project, traditional biomass fuel usage can be reduced, aiding in the mitigation of deforestation and soil erosion, thereby lowering the risk of natural disasters. These stoves produce less indoor air pollution compared to traditional methods, thereby diminishing the risk of respiratory illnesses, especially among vulnerable groups such as women and children.
Risks to stakeholder participation	Few stakeholders may be reluctant to participate in the project due to financial constraints.	The project proponent is constructing the improved cookstove with the help of Heritage foundation; hence, all the stakeholders can participate in the project without any financial burden.
Working conditions	No risk identified	Adopting improved cookstoves leads to less collection of wood from forests for cooking. With the improvement in household air pollution, the likelihood of respiratory and other pulmonary ailments is decreased. The time taken for cooking is also considerably reduced, leaving time for the villagers to take up other gainful employment.
Safety of women and girls	No risk identified	Constructing improved cookstoves as part of the project ensures the safety of women and girls, eliminating potential safety hazards.

Safety of minority and marginalized groups, including children	No risk identified	The project is the construction of improved cookstoves, which will not lead to any safety problems for minority and marginalized groups, including children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The project is to construct improved cookstoves, which will lead to a reduction in indoor air pollution. Hence, no risk has been identified.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified⁶	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	There is no discrimination during the project's current monitoring period. The same can be verified from the grievance register. Also, any such discrimination will be considered a punishable offense in the host country (Pakistan) and other developing countries, as well as LDC.
Sexual harassment	No risk identified	There were no sexual harassment cases during the project's current monitoring period. The same can be verified from the grievance register. Also, any such harassment will be considered a punishable offense in the host country (Pakistan) and other developing countries, as well as LDC. The project proponent, Core Carbon Canada, also has a stipulated HR and sexual harassment policy that is strictly followed.
Equal pay for equal work	No risk identified	PP provides equal pay for equal work

⁶ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Gender equity in labor and work	No risk identified	The project activity provides equal opportunities for women, men, or any vulnerable/ marginalized community by considering the country's laws and regulations to contribute both in volunteer and working positions. Further, the projects designed are implemented for equal participation by any gender, nationality, ethnicity, religion, and race.
Forced labor	No risk identified	The project is implemented with the full consent of the household and is voluntary and not the victims of forced labor
Child labor	No risk identified	The people involved in the project, both in volunteer and working positions, not the victims of child labor
Human trafficking	No risk identified	The project proponent recruits working staff in a very transparent manner, not the victims of human trafficking

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project activity involves constructing energy-efficient improved cookstoves to replace traditional cookstoves. The field team enters the beneficiary household with the house owner's permission. Core Carbon Canada also has a stipulated HR policy that considers participation by all demographic forms. Further, the projects designed are implemented for equal participation by any gender, nationality, ethnicity, religion, and race. Core Carbon Canada respects the rights of IPs, LCs, and customary rights holders in line with applicable international human rights law, the United Nations Declaration on the Rights of Indigenous Peoples, and ILO Convention 169 on Indigenous and Tribal Peoples.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
-------------------------	--

No risk identified	The project involves the construction of improved cookstoves for rural households. No harm is done to Indigenous people and the cultural heritage in the project location.
--------------------	--

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project activity involves the implementation of energy-efficient improved cookstoves in the participating households and does not involve resources owned by the stakeholders.
No risk identified	With prior consent, the Core Carbon Canada field team constructed the ICS only at the property owner's premises. The project proponents confirm that all property rights are recognized, respected, and supported. The project involves distributing improved cookstoves to individual households, and no land acquisition has occurred.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	In all the project activity instances under the project description, PP will provide the project devices (Energy Efficient Improved Cookstoves). Beneficiaries will be considered legal owners of the project devices. In this mechanism, energy-efficient cookstoves will be installed for beneficiaries. The ownership rights of the carbon assets generated from this project will be with Core Carbon Canada, and Core Carbon Canada will sign a detailed end-user beneficiary agreement mentioning all the criteria with all beneficiaries before the commissioning of the project devices.
Summary of the benefit sharing plan	Beneficiaries will be considered legal owners of the project devices. In this mechanism, energy-efficient cookstoves will be distributed to beneficiaries. The ownership rights of the carbon assets generated from this project will be with the project proponent, Core Carbon Canada. Core Carbon Canada has signed a detailed end-user beneficiary agreement mentioning all the criteria with all beneficiaries before commissioning the project devices. The agreement also explicitly states that the carbon rights

	are with Core Carbon Canada, and the same was explained to the beneficiaries before the installation of the improved cookstove.
Approval and dissemination of benefit sharing plan	Since the Energy Efficient Improved Cookstove is provided to users with the support of the Heritage Foundation, no benefit-sharing through carbon revenue is planned for the end users.
Benefit sharing during the monitoring period	In all the project activity instances under the project description, the project devices (Energy Efficient Improved Cookstoves) were provided by Core Carbon Canada. Beneficiaries are considered legal owners of the project devices. In this mechanism, energy-efficient cookstoves were distributed to beneficiaries. The ownership rights of the carbon assets generated from this project are with Core Carbon Canada and Core Carbon Canada signed a detailed end-user beneficiary agreement mentioning all the criteria with all beneficiaries before the commissioning of the project devices. As the Energy Efficient Improved Cookstove is provided to the end user with the help of Heritage Foundation, no benefit sharing is planned via Carbon Revenue with the end users.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project is the construction of improved cook stoves in rural households, which does not affect biodiversity and ecosystems. Therefore, no risk has been identified.
Soil degradation and soil erosion	No risk identified	The project is the construction of improved cook stoves in rural households and does not involve any activity leading to soil degradation and erosion. Therefore, no risk has been identified.
Water consumption and stress	No risk identified	The project is to construct improved cook stoves in rural households, which will not affect

		water consumption and stress. Therefore, no risk has been identified.
--	--	---

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

Species and habitat	Not Applicable
Areas needed for habitat connectivity	
...	...

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	Not Applicable	
Areas for habitat connectivity		

2.4.2 Introduction of species

Species introduced	Classification	Justification for use	Adverse effects and mitigation
Not applicable			

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
Not applicable	

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	Not applicable	

2.4.3 Ecosystem conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	Not applicable	

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0050	Energy efficiency and fuel-switch measures in cookstoves	1.0 ⁷
Tool	Tool 33	Default values for common parameters	2.0 ⁸
Guideline	-	Sampling and surveys for CDM project activities and programs of activities	04.0 ⁹
Standard	-	VCS Standard	4.7 ¹⁰
Standard	-	Sampling and surveys for CDM project activities and programs of activities	09.0 ¹¹

3.2 Applicability of Methodology

According to Methodology VM0050 version 1.0.

⁷ <https://verra.org/wp-content/uploads/2024/10/VM0050-EE-and-Fuel-Switch-Measures-in-Cookstoves-v1.0.pdf>

⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-33-v2.0.pdf>

⁹ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

¹⁰ [VCS-Standard-v4.7-FINAL-4.15.24.pdf](https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf)

¹¹ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

Methodology ID	Applicability condition	Justification of compliance
VM0050	1) The project activity corresponds to: a) Replacement of non-renewable biomass (e.g., firewood, charcoal)-fired cookstoves with any of the following: i) More efficient project devices that use the same fuel as in the baseline; ii) Efficient project devices fired by renewable biomass or bioethanol; iii) Efficient project devices fired by liquefied petroleum gas (LPG); or iv) Electric-powered project devices. b) Replacement of solid or liquid fossil fuel (e.g., coal, kerosene)-fired cookstoves with any of the following: i) Efficient project devices fired by renewable biomass or bioethanol; ii) Efficient project devices fired by LPG; or iii) Electric-powered project devices.	a) The grouped project involves the replacement of non-renewable biomass with renewable biomass. i) The grouped project device does not use the same fuel as in the baseline. Instead, the project uses wood chips and agricultural waste. ii) The grouped project device is fired by renewable biomass, i.e., wood chips and agricultural waste. iii) The grouped project is not using liquefied petroleum gas (LPG); it is replacing non-renewable biomass with renewable biomass. iv) The grouped project is not an electric-powered project. b) The grouped project is not based on solid or liquid fossil fuels. i) The grouped project device uses renewable biomass. ii) The grouped project is not an LPG-based project. iii) The grouped project is not an electric-powered project.
	2) Project devices are used in households, communities, institutions, or small or medium enterprises (SMEs), collectively referred to in this methodology as the “target population.”	The grouped project devices are used in households.
Use of renewable biomass		

	3) Where renewable biomass is used, it is exclusively renewable and qualifies as one of the following: a) A by-product, residue, or waste stream from agriculture, forestry, and related industries; b) Originating from dedicated plantations that comply with all relevant applicability conditions in the most recent version of CDM TOOL16.	a) The biomass used qualifies as a by-product, residue, or waste stream from agriculture, forestry, or related industries, such as wood chips, sawdust, or industrial biomass, and agricultural waste residues. These materials are otherwise non-utilized or discarded, requiring no additional land use or primary production, thereby meeting the criteria under Section (a). b) N/A
	4) Where biomass residues are used, they would have been left to decay or burned without energy recovery before implementation of the project activity, and their use does not involve a decrease in carbon pools – in particular of dead wood, litter, or soil organic carbon – on the land areas from which the biomass residues originate.	4) The biomass residues used are derived from sustainable sources such as agricultural waste, forest residues, or by-products of wood processing, which would have naturally decomposed or been burned without capturing energy. By utilizing these residues for energy production, the project ensures that the natural carbon cycle remains undisturbed, avoiding a decrease in carbon pools like soil organic carbon, dead wood, or forest litter.
	5) Where biomass residues from a production process are used, project implementation does not result in an increase in the processing capacity of raw input or any other substantial changes (e.g., product change) in this process.	5) The project does not involve expanding or altering the production facilities where biomass residues are generated. The biomass residues used in the project, such as wood chips or pellets, are by-products of the existing production processes, and the project does not require more raw materials to be processed than

		would have been otherwise. This ensures that the project remains within the current production capacity and does not incentivize additional raw input usage.
	6) Where more than one type of renewable biomass is used, each of the biomass types used complies with the applicability conditions.	6) All renewable biomass types used—whether derived from wood residues, agricultural waste, or other sources—meet the criteria for being considered renewable. This includes ensuring that the biomass is sourced from sustainably managed forests, agricultural processes, or other renewable sources without leading to deforestation, land degradation, or depletion of carbon pools.
	7) Where project activities introduce renewable biomass as charcoal, it is renewable charcoal produced by efficient charcoal production processes (e.g., retort sedentary kilns, improved sedentary kilns, Casamance kilns). Methane produced during the charcoaling process is captured and destroyed or combusted for energy purposes.	7) The grouped project is not based on introduction of renewable biomass as charcoal, this is replacing non-renewable biomass with renewable biomass.
	Cookstove characteristics and use	
	8) Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%.	8) Project device is using renewable biomass with multi portable, in-situ cookstove with an initial thermal efficiency of 25%
	9) Project devices using LPG or bioethanol are single-pot, multi-pot	9)The grouped project is Not based on LPG

	portable, or in-situ cookstoves with an initial thermal efficiency of at least 30%.	
	10) Electric project devices meet the maximum risk factor score of 15 on the Cookstove Durability Protocol and have the following minimum thermal efficiency: a) Hot plates and electric hobs: 40% b) Induction stoves and other electric stoves: 70%	10) The grouped project is not an electric project device
	11) Project devices using LPG comply with all of the following conditions: a) The baseline fuel either includes non-renewable biomass or is a more carbon-intensive fossil fuel (demonstrated by the baseline survey, see Section 6.2); b) The project has a provision for metering LPG supplied to each consumer at the LPG filling station, in order to determine household LPG consumption; and c) The project does not seek to issue any carbon credits for periods after 31 December 2045.	11) The grouped project is not LPG based.
	12) Electric project devices use the following electricity sources: a) Decentralized renewable energy systems: Decentralized energy systems using fossil fuels are not eligible, except for backup generators that supply less than 1% of the annual electricity of the	12) The grouped project is not based on an electric device

	decentralized renewable energy system. b) Self-generated renewable electricity (with a maximum of 20% electricity from non-renewable sources for backup); or c) National or regional electricity grid.	
	13) The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description.	13)The project proponent provides improved cookstoves (ICS) to households with the support of the Heritage Foundation.
	14) Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device.	14)The project proponent conducts continuous monitoring of improved cookstoves (ICS) to assess their condition. When repairs are needed, we provide maintenance services; otherwise, we construct new cookstoves to ensure optimal performance and sustainability.
	Avoiding harm and double counting	
	15) Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as	15)To prevent double counting of emission reductions in cookstove project, the grouped project use a unique identification number for each cookstove to identify and avoid double counting.

	geographic coordinates, complete address information).	
	16) The project complies with any national, sub-national, or local regulations or guidance for the installation, commercialization, distribution, and use of improved cookstoves and/or fuel supply and use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented.	16)In Pakistan, this project complies with national and local regulations for installing and distributing improved cookstoves (ICS), supporting air quality improvement and emissions reduction aligned with Sustainable Development Goals (SDGs). By adhering to standards from Pakistan's Environmental Protection Agency (EPA) on thermal efficiency, emissions, and safe fuel usage, the project ensures that cookstoves meet health and safety benchmarks while contributing to sustainable energy solutions.
	17) Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description.	17)The grouped project is not applicable. The project activities do not overlap with any REDD+ projects or jurisdictional REDD+ programs, and the expected fuel source area of the project is not in conflict with such initiatives. Therefore, the risk of double counting is not relevant in this case.

3.3 Project Boundary

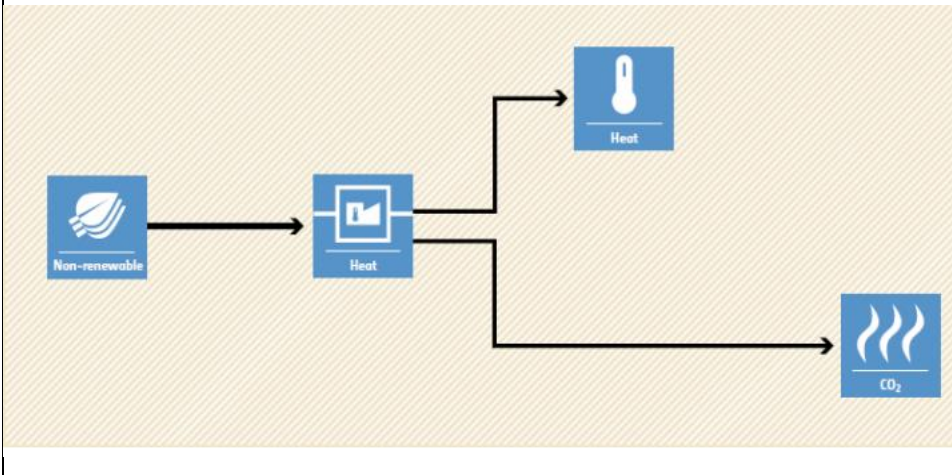
Source		Gas	Included?	Justification/Explanation
Baseline	Source 1	CO ₂	Yes	The major source of emissions.
	Emissions from use of non-renewable biomass/Fossil fuel	CH ₄	Yes	The major source of emissions.
		N ₂ O	Yes	The major source of emissions.
		Other	No	No other sources were identified.
Project	Source 1	CO ₂	Yes	The major source of emissions.
	Emissions from use of non-renewable biomass/Fossil fuel	CH ₄	Yes	The major source of emissions.
		N ₂ O	Yes	The major source of emissions.
		Other	No	No other source was identified.

ICS installation is in Sindh, Pakistan.



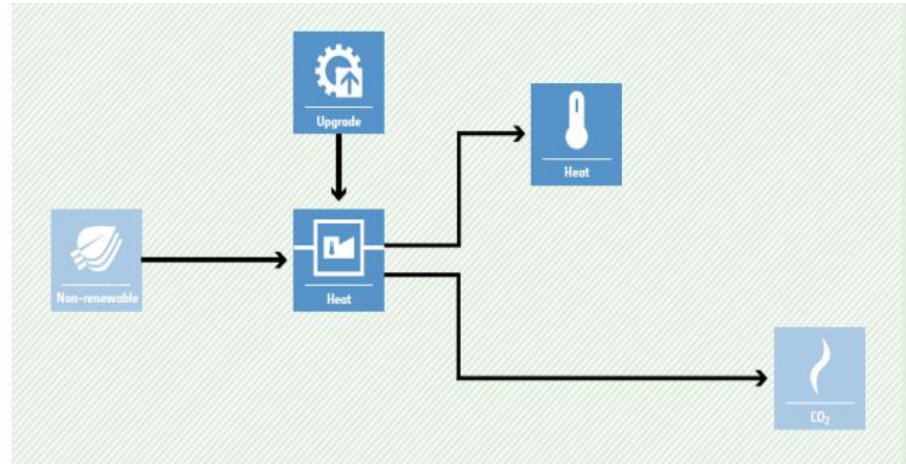
BASELINE SCENARIO

Continuation of the current situation, i.e., use of non-renewable biomass as fuel for the existing, less-efficient thermal applications



PROJECT SCENARIO

Installation of more efficient thermal energy generation units utilizing non-renewable biomass and/or complete replacement of existing less-efficient thermal applications and/or retrofitting of existing thermal energy generating appliances reduces GHG emissions by saving non-renewable biomass.



Activity Method

Step 1: Regulatory Surplus

National Climate Change Policy¹⁷ are some of the laws in Pakistan that do not explicitly mandate the use of improved cookstoves in households. While the government encourages the use of clean fuels such as LPG, many families face challenges in affording and refilling LPG cylinders due to limited income. As a result, the usage of LPG remains low, and most rural households continue to rely on conventional stoves for cooking. Notably, there is no explicit mention of improved cookstove projects in existing laws and regulations. Therefore, the implementation of such projects is a voluntary initiative by project proponents. Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by Core Carbon Canada.

Step 2: Positive List

As per Section 3.2, the project meets the applicability conditions of the methodology, which represents the positive list.

Since the project activity installs or constructs stoves for the end-users with the support of the Heritage Foundation and has no other source of revenue besides the sale of GHG credits, it should be considered additional.

The project is not implemented as part of government schemes or supported by bilateral/multilateral funds.

Conclusion: As the project fulfils the above conditions, it is deemed additional.

If any project activity instance does not follow the positive list stipulated by methodology, then additionality needs to be demonstrated as per tool requirement.

Step 3. Project Method

Step 3 is not applicable, as the project activity involves the installation of cookstoves for the end-users with the support of the Heritage Foundation and do not have any additional revenue sources apart from the sale of GHG credits.

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

¹⁷ http://www.gcisc.org.pk/National_Climate_Change_Policy_2012.pdf

☐ Yes☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes☒ No

If no, describe which mandated laws, statutes, or other regulatory frameworks require project activities and provide evidence of systematic non-enforcement to demonstrate regulatory surplus.

3.5.2 Additionality Methods

Not applicable

3.6 Methodology Deviations

The project did not apply any project deviations

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

VCS methodology VM0050, version 1.0 allows the use of a correction factor of 0.95 applied to the overall emissions reductions to account for any possible leakage. This factor has been applied to the emissions reductions presented in this report

During the current monitoring period, no incident or event occurred, that could affect GHG emissions reduction and approved monitoring plan.

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The methodology does not account for baseline emissions separately but quantifies net emission reductions the project achieves. Please refer to Section 5.4.

5.2 Project Emissions

The methodology does not account for project emissions separately but quantifies net emission reductions the project achieves. Please refer to Section 5.4.

5.3 Leakage Emissions

Leakage shall be considered a default of 0.95 per the methodology. This value is already used for the Adjustment factor. (Adj_{LE}).

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Baseline emissions are calculated as follows

$$BE_y = \sum_{i,j,k} EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO_2} \times f_{NRB,y} \times EF_{b,i,nonCO_2})$$

Where:

BE_y = Baseline emissions during year y (t CO₂e)

$EC_{i,y}$ = Average energy consumption of baseline device type i in year y (TJ)

$N_{j,k,y}$ = Number of commissioned project devices of type j from batch k in year y

$n_{j,k,y}$ = Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)

EF_{b,i,CO_2} = CO₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO₂/TJ)

$f_{NRB,y}$ = Fraction of woody biomass that is established to be non-renewable used by baseline device in year y; this variable is not considered for fossil fuels (fraction)

$EF_{b,i,nonCO_2}$ = Non-CO₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO₂e/TJ)

i = Baseline device type and its respective fuel type

j = Project device type and its respective fuel type

Average Energy Consumption of Baseline Device ($EC_{i,y}$)

The average energy consumption of baseline device type i is calculated as follows:

$$BC_{i,y} = BC_{b,i,y} \times NCV_{b,i}$$

Where:

BCb,i,y = Fuel used per baseline device type i during year y (tonnes)

NCVb,i = Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

Intake period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
15-March-2023 to 31-December-2023	224,515	0	0	224,515	0	224,515
01-January-2024 to 31-December-2024	281,609	0	0	281,609	0	281,609
01-January-2025 to 31-December-2025	281,609	0	0	281,609	0	281,609
01-January-2026 to 31-December-2026	281,609	0	0	281,609	0	281,609
01-January-2027 to 31-	281,609	0	0	281,609	0	281,609

December-2027						
01-January-2028 to 31-December-2028	281,609	0	0	281,609	0	281,609
01-January-2029 to 31-December-2029	281,609	0	0	281,609	0	281,609
01-January-2030 to 14-March-2030	55,550	0	0	55,550	0	55,550
Total	19,69,719	0	0	19,69,719	0	19,69,719

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	EF _{b,i,CO2}
Data unit	t CO ₂ e/TJ
Description	CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario
Source of data	Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	112 T CO ₂ /TJ

Justification of choice of data or description of measurement methods and procedures applied	3) Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 112 t CO ₂ /TJ
Purpose of data	Calculation of baseline and project emissions
Comments	N/A

Data / Parameter	EF _{b,i,nonCO2}
Data unit	t CO ₂ e/TJ
Description	Non-CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario
Source of data	• Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	9.46
Justification of choice of data or description of measurement methods and procedures applied	3) Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 9.46 t CO ₂ e/TJ (AR5 GWP)
Purpose of data	Calculation of baseline emissions
Comments	—

Data / Parameter	NCV _{b,i}
Data unit	TJ/tonne
Description	Net calorific value of baseline fuel used by baseline device type i

Source of data	Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC Guidelines for National Greenhouse Gas Inventories The values for wood and charcoal in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories are: Wood: 0.0156 TJ/tonne
Purpose of data	Calculation of baseline emissions
Comments	—

Data / Parameter	BCex-ante,b,i
Data unit	tonne
Description	Ex-ante annual average quantity of fuel used per baseline device type i
Source of data	Option 1: A measurement campaign following the Kitchen Performance Test Protocol must be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The campaign must achieve a confidence and precision of at least 90/10 for the target parameter of average daily fuel consumption per adult equivalent. Where the project does not achieve the target precision in a monitoring period, the project proponent must apply an appropriate conservativeness deduction as per Section 4 of the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The result must be scaled appropriately using the average household size to obtain the value of BCex-ante,b,i.
Value applied:	Option 2

	Firewood: 0.5 tonnes/capita/year of air-dried wood (from 0.0012 TJ delivered/capita/year with 0.0156 TJ/tonne NCV, and thermal efficiency of 15%)
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of data	Calculation of baseline emissions
Comments	Energy consumption must be determined once prior to validation.

6.2 Data and Parameters Monitored

Data / Parameter	$N_{j,k,y}$
Data unit	Number
Description	Number of commissioned project devices of type j from batch k in year y
Source of data	Project proponent's project database
Value applied:	50,000
Justification of choice of data or description of measurement methods and procedures applied	Established ex ante prior to start of the project activity
Purpose of data	Calculation of baseline and project emissions
Comments	The number of project devices must be recorded in a database, sales record, or similar to ensure transparency.

Data / Parameter	$n_{j,k,y}$
Data unit	Fraction

Description	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y
Source of data	Monitoring
Description of measurement methods and procedures applied	Option 2 (surveys): Based on an adoption rate determined by a survey according to the most recent version of the CDM Standard for Sampling and Surveys for Project Activities and Programmes of Activities and achieving 90/10 confidence precision for the proportion of devices in operation. The lower end of the 90% confidence interval must be used to ensure conservativeness. The adoption survey must include: a) Kitchen observation (which includes visual and physical checks of the stove and its components), including photographic evidence; and b) Interview with the primary cook. The project proponent must provide proof of training and supervision to ensure field teams have the capacity required to complete adoption surveys successfully.
Frequency of monitoring/recording	Option 2 (surveys): annually
Value applied:	
Monitoring equipment	
QA/QC procedures applied	Option 2 (surveys): Where the project device is not working or not being used at the time of conducting the survey, it should be conservatively assumed that the project device has not been active since the date on which the last adoption survey was conducted.
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	-

Data / Parameter	fNRB,y
Data unit	Fraction or %
Description	Fraction of woody biomass that is established to be non-renewable used by device in year y
Source of data	Option 3) Calculated based on CDM TOOL30 and applying an uncertainty discount of 26% ¹⁸
Value applied:	0.58
Justification of choice of data or description of measurement methods and procedures applied	Calculated on CDM TOOL30
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	BCb,i,y
Data unit	tonnes
Description	Fuel used per baseline device type i during year y
Source of data	Option 1: A measurement campaign following the Kitchen Performance Test Protocol in control households that do not participate in the project, established before validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption.

¹⁸ For example, if fNRB calculated based on TOOL30 is 0.60, the fNRB applied to emission reduction calculations is $0.60 \times (1 - 0.26) = 0.44$. The 26% uncertainty discount is derived from 100% uncertainty in fNRB, assuming a normal distribution. The discount is calculated using a standard deviation of 61% ($100\%/1.6449$) and a one-sided 66.6% confidence interval ($61\% \times 0.4307 = 26\%$). A fixed uncertainty discount is applied across all projects using TOOL30

Description of measurement methods and procedures applied	Option 1: Follow-up baseline surveys must be conducted every two years in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption. Measurement campaign must be updated where follow-up baseline surveys show that the fuels, fuel sources, or technologies used by the control group are no longer statistically equivalent to the pre-project conditions of project households. The measurement campaign must be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The result must be scaled appropriately using the average household size to obtain the value of BC_{b,i,y}.
Frequency of monitoring/recording	
Value applied:	4.818
Monitoring equipment	
QA/QC procedures applied	The campaign must achieve a confidence and precision of at least 90/10 for the target parameter of average daily fuel consumption per adult equivalent. Where the project does not achieve the target precision in a monitoring period, the project proponent must apply an appropriate conservativeness deduction as per Section 4 of the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A
Data / Parameter	BC _{p,j,k,y}

Data unit	tonnes or m3
Description	Average quantity of fuel used by project device type j from batch k during year y
Source of data	Monitoring
Description of measurement methods and procedures applied	Option 1: Kitchen Performance Test (KPT) A measurement campaign following the Kitchen Performance Test protocol must be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The campaign must achieve confidence and precision of at least 90/10 for the target parameter of average daily fuel consumption per adult equivalent. The result must be scaled appropriately using the average household size to obtain the value of BC _{p,j,k,y} . Project activities applying KPTs must also measure the quantity of fuel used by pre-project devices that are still operational and provide it in the calculation sheet and monitoring reports (this is a reporting-only requirement).
Frequency of monitoring/recording	Biennial or annual for Option 1
Value applied:	4.818
Monitoring equipment	
QA/QC procedures applied	As a crosscheck, compare results to government publications, peer-reviewed literature, third-party assessments, and/or official data or statistics. Where SUMs are used to measure project stove adoption, the stove usage indicated by the measurements for this parameter must be consistent with the frequency of use indicated by SUM measurements.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

6.3 Monitoring Plan

The PP is responsible for all monitoring procedures but cooperates with regional or local institutions involved in ICS installation. All personnel involved in the monitoring receive adequate training. Along with the surveys, monitoring also includes:

- The physical condition of the project cookstove.
- The use of baseline stoves in the project activity.

A dedicated team is responsible for monitoring the surveys. All personnel involved in the monitoring are trained to ensure that each of them undertakes an appropriate monitoring assignment according to the Monitoring Plan. The project participants maintain the database. The project participants or a representative of the project participants train the database users on how to do the data entries.

The activities involved in the project are as follows:

Activity		Responsibility
ICS installation	Installation of cookstoves to project sites.	Implementation partner
Training partners, follow-up, handholding and mentoring	Training and orientation of field staff on the use of cook stoves.	Implementation partner
	Community-level awareness and orientation meetings on cookstoves program and benefits.	Implementation partner/PP/community leaders
	Orientation of the beneficiaries for cookstove use and upkeep.	Implementation partner
	Installation of cookstoves to selected participants.	Implementation partner
Stakeholder Consultation	Physical meeting.	Implementation partner
	Stakeholder feedback round/grievance redressal.	Implementation partner
Monitoring and Reporting (Program)	Regular follow-up and reporting on usage of cookstoves.	Implementation partner
	Assessment of issues, if any, in the usage of the cookstove.	Implementation partner

The parameters mentioned in section 6.2 are monitored as part of the monitoring plan.

Project database:

A central project database is operated and maintained by the PP to ensure the completeness and accuracy of monitoring information. The project database is updated regularly as new data becomes available. Original copies of the installation forms are kept and maintained.

PP gathers the necessary information to identify households using its ICS during the project. Each ICS is assigned a unique serial number to facilitate this process. This number is recorded during the registration process along with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of construction/installation
- ICS serial number

The information collected is stored in the electronic database, which serves as a project database for project monitoring and sampling purposes.

Operation & Maintenance:

Our field staff regularly visit to ensure everything functions appropriately during the Improved Cookstove (ICS) operation and maintenance. If any issues are identified during these visits, we guarantee they will be resolved within a week. This proactive approach helps us ensure that the ICS continues to operate efficiently, providing reliable cooking solutions for our beneficiaries.

QA/QC process:

PP applies measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The data collected and recorded in the project database is checked regularly by PP as part of the quality control process to check for any discrepancies or missing information.

All information regarding Pakistan Chulah installation is documented through beneficiary and agreement forms in the project database. Due to the large number of Pakistan Chulah installed, monitoring the project Chulah consists of checking a representative sample annually through project surveys to ensure that project Chulah is still operating in the monitoring period. A questionnaire for the survey is prepared per the requirements mentioned in the VM0050 - Methodology for Energy efficiency and fuel-switch measures in cookstoves, Version 1.0.

Sampling Approach Used:

PP has adopted a simple random sampling approach to select households. Simple random sampling is chosen due to its conceptual straightforwardness, ease of implementation, and appropriateness for relatively homogeneous populations.

In our context, the rural households within each PAI have similar socio-economic and environmental characteristics, making them relatively homogeneous regarding fuelwood consumption patterns. It supports that the primary selection criterion for installation is the need for ICS, indicating a shared need and context among the target households.

To perform simple random sampling effectively, a comprehensive list of all households that receive ICS in the PAI serves as the sampling frame. This sampling frame ensures that each household has an equal probability of being selected for the survey.

Precision Level:

As per the representative sampling requirements of the VM0050 -Energy efficiency and fuel-switch measures in cookstoves, Version 1.0, the project proponent conducts sample surveys for the PAI to achieve a confidence/precision level of 90/10 in line with the latest version of the "Standard for sampling and surveys for project activities and programme of activities", Version 9.0.

Sample Size:

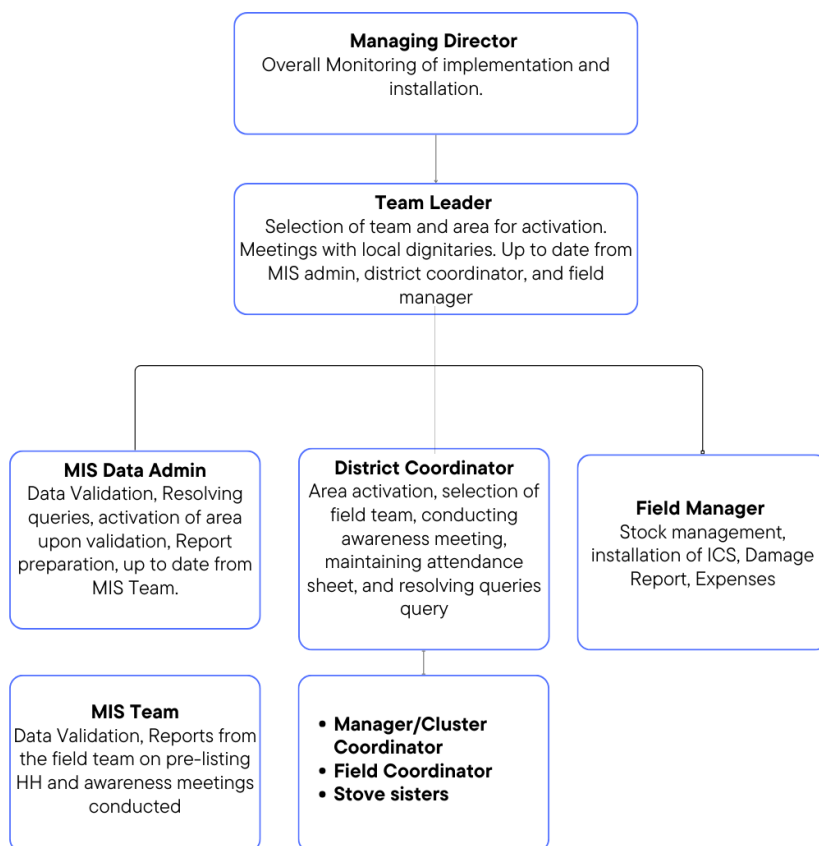
Using the CDM sample size calculator of "Standard for sampling and surveys for project activities and programme of activities", Version 9.0., the minimum number of samples required for achieving the confidence/precision level of 90/10 shall be determined for the PAI.

Frequency of Measurement:

This simple random sampling was done once in the baseline survey to estimate the fuelwood consumption for the PAI. Project monitoring surveys shall be conducted at least once every two years using the same methodology.

QA/QC Procedure:

The PAI sample size has been calculated using CDM Guidelines "Sampling and surveys for CDM project activities and programmes of activities", Version 4.0



7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$N_{j,k,y}$
Data unit	Number
Description	Number of commissioned project devices of type j from batch k in year y

Value applied:	57,000
Comments	-

Data / Parameter	$NCV_{b,i}$
Data unit	TJ/tonne
Description	Net calorific value of baseline fuel used by baseline device type i
Value applied:	0.0156
Comments	-

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass that is established to be non-renewable used by the device in year y
Value applied:	0.58
Comments	-

7.2 Baseline Emissions

Mentioned in section 5.1

7.3 Project Emissions

Mentioned in section 5.2

7.4 Leakage Emissions

Mentioned in section 5.3

7.5 GHG Emission Reductions and Carbon Dioxide Removals

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
15-March-2023 to 31-December-2023	224,515	159,406	-29	The difference is due to the number of cookstoves operation due to distribution over the year.
01-January-2024 to 31-December-2024	281,609	214,022	-24	The difference is due to the number of cookstoves operation due to distribution over the year.
01-January-2025 to 31-October-2025	55,550	45,551	-18	The difference is due to the number of cookstoves operation due to distribution over the year.
Total	5,61,674	418979	-23.66	The difference is due to the number of cookstoves operation due to distribution over the year.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.